

Autonomous Theory Construction Laboratory

Lecture 3: Grammar of Candidate Theories

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Overview

A typed grammar constrains candidate expressions before dimensional and symmetry checks are applied.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$G ::= \Phi | \partial G | G \cdot G | f(G; \theta)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$\dim(\text{term}) = \dim(\text{target})$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$\text{gate}(x) = 1/(1 + e^{-(x-x_0)/w})$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Yun, ch. 3
- Candidate Grammar 26F-2
- Canonicalization Note CN-8